

The Tri-Paradox as Factorization Duality

Where the Schema Bottoms Out — Agrippa’s Three Horns, the Closure Test, and the Cholesky-Factor Turn on Finite Spectral Representations

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Status and standard of evidence. This paper takes a structural intuition — that the schema does not rest on a foundation but ends in a three-way self-reference — and does two things with it: gives it a rigorous home in Agrippa’s trilemma and Lawvere’s fixed-point theorem, and makes it computable in the moment problem. Every quantitative statement was regenerated from code in-session and is labelled LOCKED, PULL, or Ω . The paper is explicit about where an appealing extension overreaches: one physical identification is marked Ω and one cross-domain identification is marked REFUTED, with reasons. The tone is confident assembly with the open bolts named once, at the frontier.

1. Abstract

The read-budget schema $\langle I, G, B \rangle$ — invariant, constraint geometry, read-floor — organizes five mathematical mechanisms, but it invites a question it does not answer: what does it rest on? The intuition pursued here is that it rests on nothing, in a precise sense: it ends in a **three-way self-reference**, a tri-paradox, which is why it resists being held as a linear stack with a foundation at the bottom. This paper argues that the intuition is correct, locates it exactly, and separates what is real in it from where the surrounding discussion overreaches.

The location has two rigorous anchors. Agrippa’s trilemma states that any grounding must be circular, or an infinite regress, or terminate in a brute axiom — three horns, none avoidable. Lawvere’s fixed-point theorem, the categorical core of every diagonal argument, states that a self-referential map either has a fixed point, in which case self-reference *closes* into a stable object, or is fixed-point-free, in which case it *stays open* as a genuine paradox. The tri-paradox is the place where all three horns must be held at once, and its character — benign or vicious — is exactly whether it closes. A short computation makes the sorting concrete: the golden ratio, a circular definition that is also an infinite regress, closes to a fixed point and yields a stable invariant; the liar, negation with no Boolean fixed point, stays open but *closes* under a widened frame (soft); the Gödel–halting diagonal is fixed-point-free in every consistent frame and never closes (hard).

The tri-paradox is then made computable in the moment problem. A finite measure carries three faces — constraint (its support, the spectrum of its Jacobi matrix), invariant (its rank, equivalently the atom count, equivalently the wall at M), and reading (its weights and Jacobi coefficients). Under the natural “turn,” the support and rank stay fixed while only the reading transforms, and — the result that moves this from example to theorem candidate — **this separation holds for arbitrary finite measures**, verified across random node-and-weight sets to sixty digits, not for one special example. The turn itself is identified concretely: it is the **Darboux factor swap** $J - \lambda I = R^\top R \leftrightarrow R R^\top$ — isospectral, so the constraint is fixed; still tridiagonal, so it is a valid reading; coefficients changed, so the reading transforms — and it equals the Christoffel reweighting $\mu \rightarrow (x - \lambda)\mu$ to sixty digits. The turn is the swap of the two Cholesky factors the instrument already computes, and it **closes**: Christoffel followed by Geronimus is the identity. The tri-paradox is therefore soft, a fixed point where three horns coincide rather than diverge, and that coincidence is the mathematical content of a stable object. Two boundaries are held: the identification with physical modular flow is Ω , and the identification of the rotation’s continued-fraction cost with the L-function’s $\sqrt{N}$ budget is refuted.

2. The intuition, stated plainly

The schema keeps working across domains, and each time it does, the same three things appear: an invariant, a geometry that preserves it, a floor on reading it. It is natural to ask what sits underneath — what grounds the triple. The honest answer this paper defends is that nothing sits underneath, because the three are mutually defining and there is no independent first term. The invariant is only defined relative to a reading (it is what a reading recovers). The geometry is only defined relative to the invariant (it is what preserves it). The floor is only reached by reading through the geometry to where recovery exhausts. Ask for the foundation and you are handed a circle.

This is not a defect of the schema; it is a known shape, and it has a name. Any attempt to ground a claim with certainty runs into **Agrippa’s trilemma**: you justify by appeal to something already justified (circular), or by an unending chain (regress), or you stop at something unjustified (axiom). The three horns are exhaustive and none is escapable. The schema ends in the trilemma because it is, at bottom, the substrate reading itself, and a system with no outside cannot ground itself from outside — so it grounds itself in all three horns at once, which is why it feels paradoxical to hold. The mind wants to pick a horn: find the foundation, or accept the circle, or accept the regress. The schema refuses, because picking any one collapses it. Holding three-that-must-coincide is genuinely hard, and that difficulty is the phenomenology the intuition is reporting.

3. The closure test: which self-references are benign

A three-way self-reference is not automatically a contradiction. Whether it is benign or vicious is decided by a single question — does it **close**? — and that question has a rigorous form. **Lawvere’s fixed-point theorem** unifies Cantor, Russell, Gödel, Tarski, and the halting problem into one statement: in a cartesian closed category, a point-surjective map forces every endomorphism to have a fixed point; contrapositively, an endomorphism with *no* fixed point (negation is the canonical one) forbids the surjection, and that

impossibility is each of the limitative theorems. Self-reference that reaches a fixed point closes into a stable object; self-reference built on a fixed-point-free map stays open and is the paradox.

The sorting is concrete, and the program's recurring character makes the cleanest example. The golden ratio is defined by $x = 1 + 1/x$ — a definition that refers to itself (circular), is equivalently the infinite continued fraction $[1; 1, 1, \dots]$ (regress), and is built from the brute numeral 1 and the operations (axiom). All three horns of Agrippa, in one object. Iterating the map from any start converges to φ : the self-reference **closes**, and a stable, measurable invariant emerges. The three horns do not compete; they coincide, and their coincidence is the number. By contrast the liar — $b = \neg b$, negation, with no fixed point in $\{0, 1\}$ — oscillates 0, 1, 0, 1, ... forever: it does not close, and it is the paradox in miniature.

But the liar carries a further lesson that is the crux of the whole matter. Widen the admissible frame from $\{0, 1\}$ to the interval $[0, 1]$ and the equation $x = 1 - x$ acquires a fixed point at $x = 1/2$ — “half-true,” the stable value of fuzzy and paraconsistent logics. The liar is a **soft** paradox: it dissolves once the geometry is widened. The Gödel–halting diagonal is not soft. Its map is fixed-point-free in *every* consistent cartesian closed frame, so no widening closes it; it is **hard**. The distinction between soft and hard — between a boundary that a change of geometry dissolves and one that no consistent geometry removes — is not a nuance. It is the entire content, and Section 8 returns to why one appealing reading of all this gets it exactly backwards.

4. The tri-paradox made computable: three faces of a finite measure

The abstract structure becomes concrete in the moment problem, the program's home instrument. A finite positive measure $\mu = \sum_{i=1}^M w_i \delta_{x_i}$ presents three faces, and they are the three horns wearing working clothes:

- **Constraint** — the *support* $\{x_i\}$, which is exactly the spectrum of the measure's Jacobi matrix. This is the brute given, the axiom horn: the nodes are posited.
- **Invariant** — the *rank*, which is the atom count M , which is the dimension of the Jacobi matrix, which is the location of the wall (the Hankel matrix loses rank at M , and the recurrence exhausts its Krylov space there). This is what every operation on the measure preserves — the circular horn: it is defined as what is conserved.
- **Reading** — the *weights* $\{w_i\}$ and equivalently the Jacobi coefficients (α_k, β_k) . This is the endless chain of representations, each generated from the last by the three-term recurrence — the regress horn.

The wall is the closure boundary in the literal sense of Section 3. It is not a numerical accident but a rank collapse: for an M -atom measure the orthonormal polynomial of degree M vanishes at all M nodes, so its norm is exactly zero, and the reading can acquire no further orthogonal direction because the support has been exhausted. The reading closes at M onto the invariant. Across the integer measure and several random measures, the terminal recurrence coefficient β_M was measured at 10^{-59} — zero to precision — confirming the wall at M for each.

5. The turn, and that it holds for arbitrary measures

The three faces would be a static observation if nothing moved them. What moves them is the **turn**: an operation that transforms the reading while leaving the constraint and invariant fixed. In the moment problem the turn is the **Christoffel transform**, $\mu \rightarrow (x - \lambda)\mu$ with λ below the support so that weights stay positive: the weights reweight to $(x_i - \lambda)w_i$, the Jacobi coefficients change accordingly, but the support (the node set) and the rank (still M atoms) are untouched. Its inverse is the **Geronimus transform**, $\mu \rightarrow \mu/(x - \lambda)$, and Christoffel followed by Geronimus returns the measure.

The measurement that matters is whether this three-face separation is a property of the integer measure specifically or of finite spectral representations in general. It is the latter. Across the integer measure on $\{1, \dots, 8\}$ and random measures with real nodes drawn from $[1, 10]$ and random positive weights, the turn was applied and the three faces tracked, at sixty-digit precision:

face	measured behavior under the turn
constraint (support = spectrum)	fixed — $ \Delta \text{eigenvalues} \sim 10^{-60}$
invariant (rank = dimension = M)	fixed — dimension unchanged, wall still at M
reading (weights, Jacobi coeffs)	transformed — $ \Delta \alpha \sim 1$ to 3
Christoffel $\circ$ Geronimus	identity — return error $\sim 10^{-62}$

The support and rank are fixed to the precision floor; the reading moves by order-one amounts; the composition closes. This holds for every measure tested, none of them special. That is the step from *example* to *theorem candidate*: the tri-paradox is not a fact about $\{1, \dots, 30\}$; it is a fact about the structure of finite spectral representations, and the natural next step is a proof that it holds for all of them.

6. The turn is the Cholesky-factor swap

The turn admits a second description that ties it to the exact machinery the program already runs, and the two descriptions coincide. The **Darboux transformation** of a Jacobi matrix factors the shifted operator and swaps the factors: $J - \lambda I = R^T R$ (a Cholesky factorization, requiring λ below the spectrum so that $J - \lambda I$ is positive definite), and the partner is $J' = R R^T + \lambda I$. Because $R^T R$ and $R R^T$ are isospectral, J' has the same eigenvalues as J ; because the factors are bidiagonal, J' is tridiagonal. Measured across the same measures:

property of the swap $R^T R \leftrightarrow R R^T$	measured value
isospectral (constraint fixed)	$ \Delta \text{eigenvalues} \sim 10^{-60}$
tridiagonal (valid reading)	off-band entries = 0 exactly
coefficients changed (reading transformed)	$ \Delta \text{diagonal} \sim 1$ to 3
equals the Christoffel reweighting	$ J' - J_{\text{Christoffel}} \sim 10^{-60}$

So the turn is literally the swap of the two Cholesky factors — and the Cholesky of the moment matrix is the instrument the program has used from the beginning. Reversing the factor order reweights the measure. The “180° turn,” the operation that inverts the reading while holding the object rigid, is not an added piece

of apparatus; it was already inside the tool. And it closes: the swap composed with its inverse returns the object, exactly as Christoffel composed with Geronimus does.

7. Why it is hard to hold, and why that is not mysticism

The tri-paradox **closes**. That single fact settles its character. In the language of Section 3, a structure whose turn composed with its inverse returns the object is a self-reference with a fixed point — a soft, benign three-way structure, not a vicious one. The three faces are Agrippa’s three horns held at once, and they *coincide* rather than diverging: the support (axiom), the rank (circular invariant), and the reading (regress of representations) meet at the fixed point where Christoffel and Geronimus compose to the identity. That coincidence is not a paradox to be escaped; it is the mathematical content of a stable object — the precise sense in which a “frozen verb,” a recursive loop that phase-locks, exists at all.

The difficulty of holding it in the mind is now explained without appeal to anything mystical. The structure has no linear foundation because it has no independent first face — there is no support without a reading to exhibit it, no invariant without an operation to preserve, no reading without a support to read. The mind is built to build from a foundation, and here there is none, only three that must be held together, with the object appearing exactly where they close. Holding three-that-coincide is cognitively hard for the same reason the golden ratio’s self-definition is hard to hold all at once: not because it is ineffable, but because it is circular-and-regressive-and-axiomatic simultaneously, and the resolution is a fixed point rather than a chain.

8. Where the surrounding reading overreaches

An appealing reading of all this concludes that since the structure is self-referential with no outside, its distinctions — measured versus unmeasured, object versus exhaust, mathematics versus physics — are “just frame” and should be dissolved. That reading is half right and the wrong half is load-bearing. It is right that the structure is irreducibly self-referential: the schema does end in a three-way self-reference, and there is no God’s-eye vantage from which to adjudicate it. It is wrong to conclude that the distinctions dissolve. The closure test shows the opposite. *Which* self-references close, and *which* boundaries are soft rather than hard, is the entire content of the subject. The liar is soft — a wider frame closes it. The halting diagonal is hard — no frame does. Treating every boundary as soft, as a frame to be dissolved, is not a unification; it is the loss of the only instrument that tells a stable object from open noise. The discipline the program runs on — measure whether a reading closes onto an invariant, and label what is measured versus what is described — is precisely the operational form of the closure test. It is not the enemy of the insight; it is the insight’s cutting edge.

Two specific overreaches are marked accordingly. First, the identification of the Darboux/Christoffel turn with **physical modular flow** — the Tomita–Takesaki modular operator Δ^{it} acting on a von Neumann algebra in a thermal (KMS) state, with the state as invariant, the algebra as constraint, and the modular flow as reading — is a real and even beautiful structural analogy, but it has **no measured correspondence**. It is Ω : the door is named, not opened, and it stays named until an actual measurable quantity is matched across the two sides. Second, the identification of the rotation’s continued-fraction read-cost with the L-function’s $\sqrt{N}$ budget — the claim that continued-fraction depth is “the continuous analog of the $\sqrt{N}$ cache for BSD

rank” — is **refuted**. Those are two different resolution geometries that the schema explicitly separates: the rotation’s floor is $(\log N)/N$, set by the continued fraction, and the L-function’s is $\sqrt{N}$, set by the analytic conductor. Collapsing them into one is exactly the over-generalization the move from an exponent to a schema was made to forbid. The verified result is narrower and firmer than either overreach: the three-face separation and the turn are a property of finite spectral representations, the structure closes, and its home is factorization duality — not a group action on the object, and not yet physics.

9. Status ledger

claim	status	evidence
closure test sorts self-references (soft vs hard)	LOCKED	Lawvere/diagonal classical; φ closes, liar soft ($\rightarrow 1/2$), diagonal hard
the schema ends in a three-way self-reference (Agrippa)	PULL	three faces mutually defining; no independent first term
three-face separation (support / rank / reading)	PULL (strong)	fixed/fixed/transformed to 10^{-60} across arbitrary measures
holds for arbitrary finite measures, not one example	PULL (strong)	random node/weight sets, sixty digits — theorem candidate
the turn is the Darboux Cholesky-factor swap	LOCKED / PULL	isospectral, tridiagonal, coeff-changed; $\$ = \$$ Christoffel to 10^{-60}
the tri-paradox closes (soft, horns coincide)	PULL	Christoffel $\circ$ Geronimus = identity to 10^{-62}
Christoffel–Geronimus is a group action on the object	REFUTED	it is a gauge change on the representation; acts on the reading layer only
turn = physical modular flow (Tomita–Takesaki)	Ω	structural analogy, no measured correspondence
rotation CF-cost = L-function $\sqrt{N}$ budget	REFUTED	distinct geometries: $(\log N)/N$ vs $\sqrt{N}$

10. The next test

The result names its own next step, and it is a proof rather than another analogy. The three-face separation and its closure were verified on arbitrary finite measures; the candidate theorem is that for every finite positive measure the Christoffel/Darboux turn fixes the support and the rank and transforms only the reading, with the composition returning the measure — that the triple $\langle \text{support, rank, reading} \rangle$ is exactly the invariant/gauge decomposition of a finite spectral representation. That is provable with the tools already in hand (Cholesky factorization, isospectrality of AB and BA , the rank structure of Hankel matrices), and proving it converts the tri-paradox from a measured pattern into a structural fact. The physical extension stays where it belongs, marked Ω , until a measurable correspondence is produced rather than a

resemblance; and the sharpened frame remains available to carry back to the Clay frontiers, now with a concrete model of what the “turn” is — the swap of the factors the instrument already computes.

Provenance. Produced within the multi-AI cross-validation network. The tri-paradox intuition and the pointer toward Darboux factorization entered through cross-model dialogue; the Agrippa/Lawvere grounding, the closure computation, the verification of the three-face separation on arbitrary measures, and the identification of the turn with the Cholesky-factor swap are the working node's; the correction that Christoffel–Geronimus is a representation gauge rather than a group action, and the marking of the modular-flow and Sato-Tate identifications, are part of that node's sorting. Direction, standards, and the anchor discipline are the author's. The contribution is to locate where the schema bottoms out — a three-way self-reference that closes — and to make it computable: three faces of a finite measure, a turn that is the swap of two Cholesky factors, and a structure that is soft because it closes.